

Ultrasonic Water Meter Manual

XKFL-5 Series



Executive standard:

Production standard: National Standard of the People's Republic of China

GB/T778.1~778.5-2018 "Drinking Cold Water Meter and Hot Water Meter"

National Standard of the People's Republic of China CJ/T434-2013 "Ultrasonic Water Meter"

Verification standard: National Metrology Verification Regulations of the People's Republic of China JJG162-2019 "Cold Water surface"

Important hint:

- (1). Please read the user manual carefully before use, and keep the user manual properly for reference;
- (2). This product is a precision measuring instrument. During installation and use, keep away from interference sources as far as possible, choose a reasonable installation location, and avoid severe impacts and collisions;
- (3). This product has undergone strict quality inspection before leaving the factory, non-standard operation or unauthorized disassembly is strictly prohibited, otherwise the consequences will be at your own risk;
- (4). The displayed patterns and prompts are window operation prompts and cannot be used as the basis for product verification;
- (5). For product verification, please choose a regular flow verification device that meets the national measurement requirements, and operate in strict accordance with the specifications;
- (6). Since the water meter will be upgraded from time to time, if the actual function of the product is inconsistent with this manual, please refer to the actual product or consult the manufacturer;
- (7). When an error message appears, please contact the manufacturer or dealer in time, otherwise it will cause traffic loss.

This product does not support reverse flow measurement, please be informed!

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1、Product Features:

- It is specially used for accurate metering of small-diameter water in residential areas or other places;
- Ultrasonic measurement, effectively avoiding the interference of traditional magnets;
- The minimum measurable flow velocity is only 0.005m/s (DN15), truly realizing drip metering;
- LCD display, complete prompt function, convenient for inquiry and diagnosis;
- Small size and low power consumption design, the actual service life is more than 8 years;
- Excellent linearity, consistent linearity within the full scale range, no need for multi-point calibration;
- Perfect software expansion function, which can meet the customization requirements of different users;
- A variety of measurement units are available to meet the needs of users in different countries;

2、Products Size (mm)

Model	Length	Width	Height
DN15	110	101	92
DN20	130	101	92
DN25	225	101	92

3、Product Specifications

No.	Parameter Item	Specification
1	Applicable Pipe Diameter	DN15、DN20、DN25
2	Voltage	DC 3 ~ 10V(typical value 3.6V)
3	Transducer frequency	1MHZ
4	Range Ratio	250/ 400

5	Water meter accuracy	2
6	Communication method	LoRa/LoRaWan /GPRS/4G/RS485/M-BUS
7	Temperature class	T30/T50
8	Maximum allowable working pressure	1.6Mpa
9	Working power	3.6V Lithium Battery
10	Degree of protection	IP68

4、Flow Range

Diameter	R	Q1	Q2	Q3	Q4
DN15	R400 (R250)	0.006m ³ /h	0.010m ³ /h	2.5m ³ /h	3.125m ³ /h
DN20	R400 (R250)	0.010m ³ /h	0.016m ³ /h	4m ³ /h	5m ³ /h
DN25	R400 (R250)	0.016m ³ /h	0.025m ³ /h	6.3m ³ /h	7.875m ³ /h

5、Installation :

5.1 installation conditions

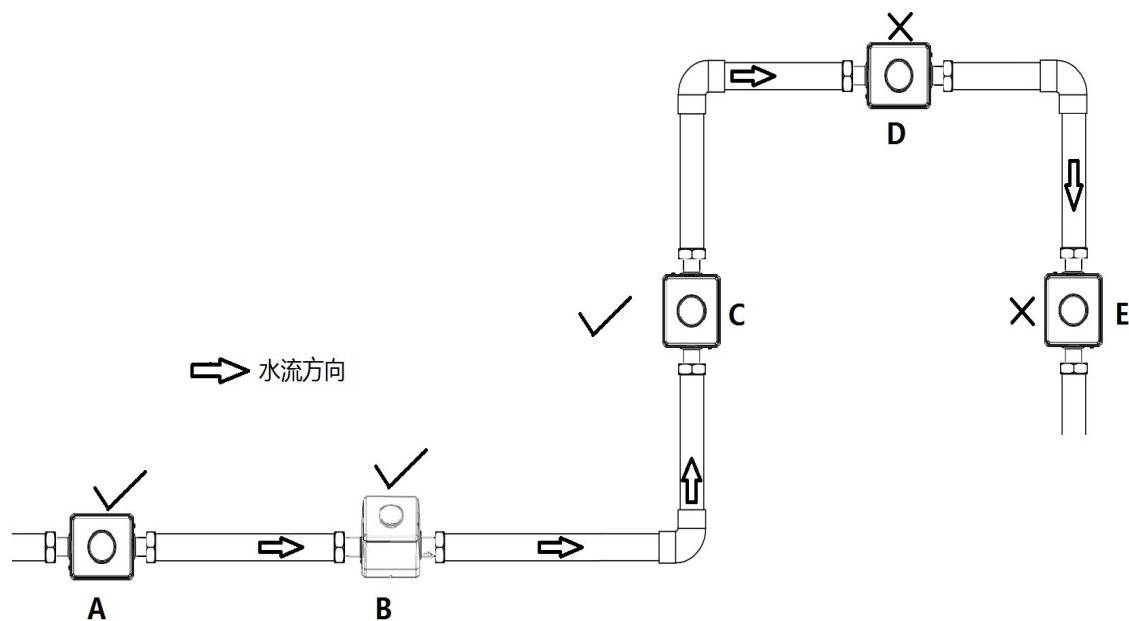
The water meter specification should be selected according to the water flow conditions of the pipeline, and the water meter caliber cannot be determined simply by the caliber of the pipeline. The installation site of the water meter should prevent exposure to the sun and freezing, and at the same time avoid pollution and flooding to prevent dirt from entering the inside of the water meter. Before installing the water meter, the pipeline should be rinsed clean, and the stones, sediment and other debris in the pipeline should be removed before installing the water meter, and the sealing cover installed at the water inlet and outlet of the water meter should be removed.

The water flow direction after the water meter is installed is indicated by the arrow on the meter body. If the water meter is installed in the opposite direction, no measurement will be performed. The pipe joints connected to the water meter should at least meet the compressive strength of MAP16, and the thread should be partially exposed during installation to avoid damage to the meter body caused by shrinkage and expansion. The connecting pipe must be installed parallel to the water meter.

It is recommended to use EPDM sealing ring when installing T30 water meter, and install PTFE sealing ring when installing T50 water meter, and the sealing ring must be brand new. The installation environment of the water meter should avoid direct sunlight, and keep away from acid, alkali, flammable, explosive and corrosive substances, and prevent strong electromagnetic field interference.

5.2 installation angle

Ultrasonic smart water meters can work normally when installed at any angle, but inclined installation will lead to changes in measurement accuracy. It is recommended to use horizontal and vertical installation directions. The correct installation method is recommended as follows:



A: Horizontal side 90 degree installation: recommended installation method

B: Installation at 45 degrees on the horizontal side: recommended installation method

C: Vertical installation of the water meter from bottom to top: recommended installation method

D: Horizontal installation at the top of the pipe network: not recommended installation method

E: Vertical installation of the water meter from top to bottom: not recommended installation method

5.3 Straight pipe requirements

Under normal circumstances, the water meter can meet the measurement accuracy requirements without additional installation of the front and rear straight pipe sections. However, it is necessary to install the front straight pipe section of the water meter in the case of severe water flow disturbance. The length of the front straight pipe section should meet the requirements of the flow profile sensitivity level U10.

5.4 work pressure

The static pressure in front of the water meter should not be lower than 0.03MPa

5.5 Installation Notes:

- (1). The diameter of the water meter should be selected according to the diameter of the pipeline at the installation site and the flow rate should be less than or equal to the usual flow rate;
- (2). The installation location should be selected where it is easy to disassemble and read the power data of the water meter;
- (3). During installation, sand, gravel, tape, iron wire and other sundries should be prevented from entering the water meter pipeline, which may cause water meter measurement failure;
- (4). When installing, the direction of water flow should be consistent with the direction of the water meter arrow;
- (5). The water meter is not suitable for measuring hot water, please do not install it in hot water places such as solar energy and bathing;

(6). After installation, please switch to the signal display interface and read the signal value. The signal value should be more than one grid.

6、 Window Operation

The water meter is usually in a low-power off-screen state, and it can be woken up when the magnet is placed in the lower left corner of the screen (where the CPA logo is located). The magnet moves across the lower left corner of the screen to realize switching and cycling of the display window.

7、 Window Display

Display order:

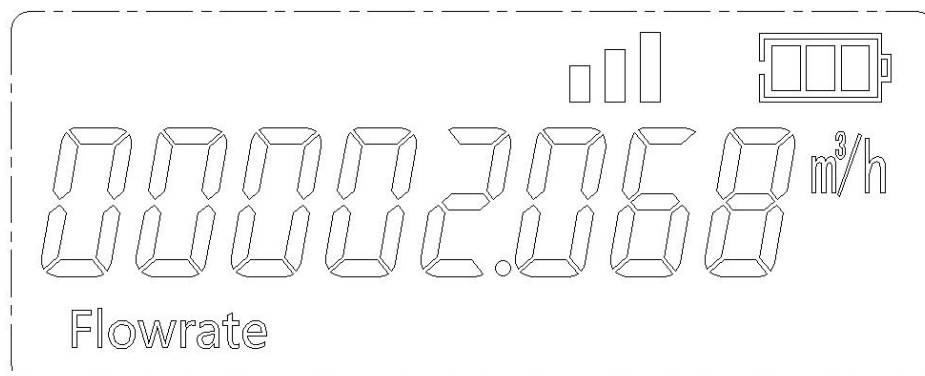
Cumulative flow $\Rightarrow$ Instantaneous flow $\Rightarrow$ Time $\Rightarrow$
Meter Number $\Rightarrow$ All screens on $\Rightarrow$ all screens off

1) Empty section alarm



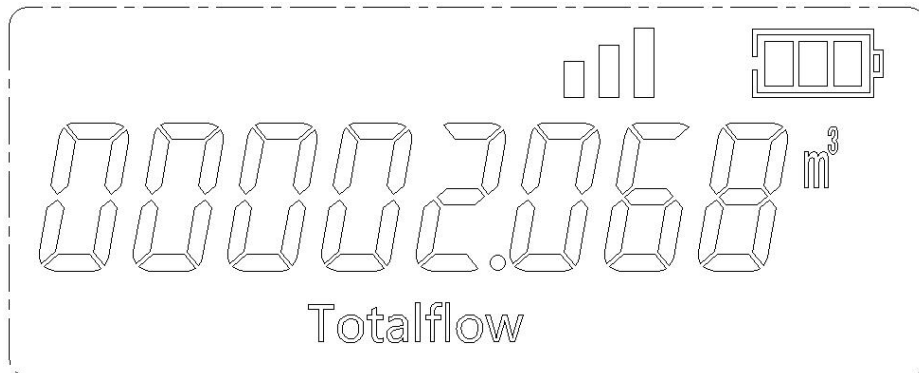
When the CPU detects the state of an empty pipe section, the icon is on, and when the pipe section is full of water, the icon disappears automatically.

2) Instantaneous flow



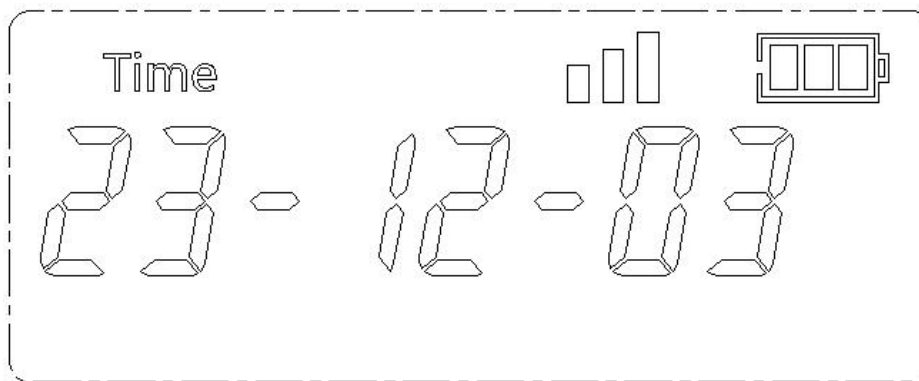
Instantaneous flow, indicating the current instantaneous flow rate of the water meter, displaying 8 digits, accurate to three decimal places, and the unit is m³/h

3) Cumulative flow



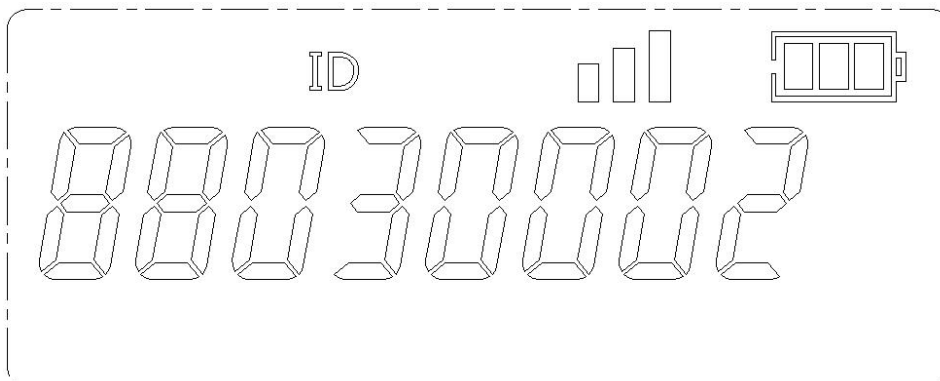
Cumulative flow, indicating the current cumulative flow value of the water meter, displaying 8 digits, accurate to 3 digits after the decimal point, and the unit is m3

4) Time



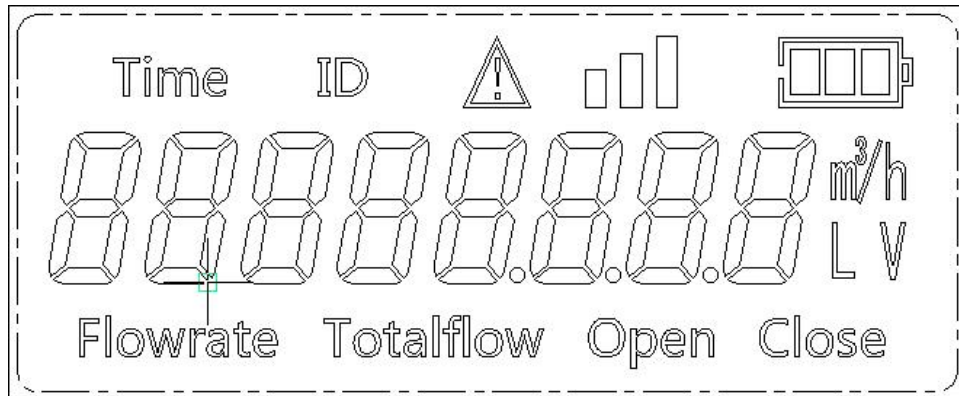
Real-time display of the current system time, the display format is year-month-day

5) Meter Number



Indicates the factory serial number of the current water meter, 8-digit display, for communication and meter reading

6) Full display

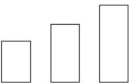


All the screens are on, which is used to identify whether the screen is in good condition.

7) Screen off

The screen is all off, which is used to identify whether the screen is in good condition.

8、 Common faults and troubleshooting methods

Fault icon	Phenomenon	Fault content	Method of exclusion
	Icon lights up	Empty pipe section alarm: there is no water or large air bubbles in the pipe	fill a pipe with water, or make a stream of water flow
	icon flashing	Not enough power	Replace battery
recharge	"Recharge" Icon lights up	Insufficient balance in the prepaid water meter	Please contact the relevant unit to recharge
	"Signal" icon None	Poor network signal	Change the installation location of the water meter, or use an external antenna

9、Packing list and accessories

Serial	content	quantity
1	Ultrasonic water meter	1set
2	Union nut and sealing ring	1pair
3	manual	1
4	certificate	1
5	Packing box and cushion	1

10、Appendix Verification method

1. The verification method follows the national measurement verification regulations JJG162-2019

"Verification Regulations for Drinking Cold Water Meters"

2. When calibrating, please choose a formal flow verification device certified by the metrology department and operate it according to the specifications of professionals;

3. This series of products (DN15, DN20, DN25) are more suitable for "start-stop method" calibration;

4. This series of products can be marked and used immediately, without additional operations, and the verification accuracy is consistent with the actual use accuracy.

5. Verification process:

1) Clamp the pipe section on the meter calibration platform, and after the air is cleared, switch the display window to the cumulative flow interface. Place a magnet for more than 6S under the cumulative flow window (the position is at the CPA mark in the lower left corner of the screen), remove the magnet, and change the cumulative flow unit from m3 to L, and the minimum resolution is 0.001L.

2) Record the initial value and final value, and calculate the error after the test is completed.

3) After entering the verification mode, the meter remains in the verification state for 2 hours. After 2 hours, it will automatically exit the verification mode. If you want to exit the verification mode during the verification process, you must place the magnet at the CPA mark for more than 6 seconds. Exit test mode.

Main components and parameters

Serial	Name	Model	Key performance indicators
1	Brass pipe section	DN15	Brass material, built-in reflection bracket and reflection sheet, $\Delta p63$, 1.6MPa
2	Brass pipe section	DN20	Brass material, built-in reflection bracket and reflection sheet, $\Delta p63$, 1.6MPa
3	Brass pipe section	DN25	Brass material, built-in reflection bracket and reflection sheet, $\Delta p63$, 1.6MPa
4	Transducer	WK-21B	1MHZ, 21mm
5	Shell	GN-FL3C	ABS
6	Circuit board	GN-FLNB	DC3.6V Power supply, broken code LCD screen display, Communication LORAWAN
7	Battery	26500	DC3.6V, 6000mAH

Main technical indicators of water meter

Battery	10 years	Protection	IP68
Caliber	DN15、DN20、DN25	Max pressure	1.6MPa
Metrology grade	2	Relative humidity	93%
Pressure loss	$\Delta p63$	Temperature class	T30/T50
Environmental level	O	Electromagnetic environment	E1
Minimum reading	0.001m ³	Maximum reading	999999.999m ³
Communication	Lorawan	Static current	$\leq 40\mu A$
Flow Field Sensitivity Level	U10/D5	Environmental level	B